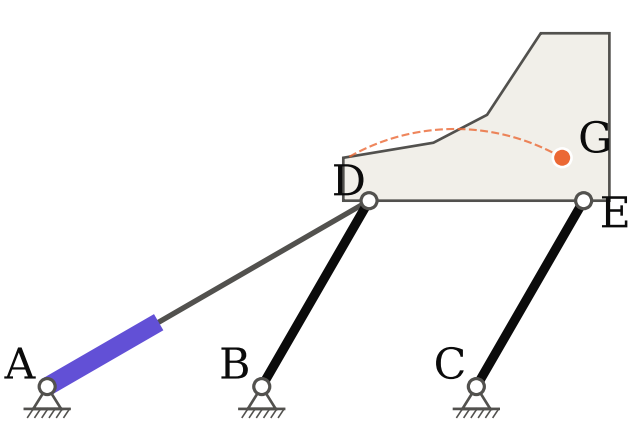
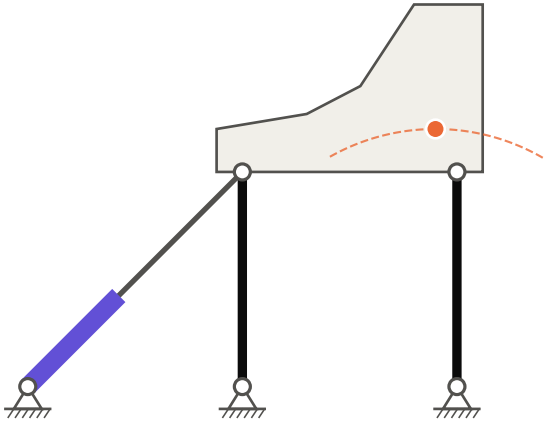
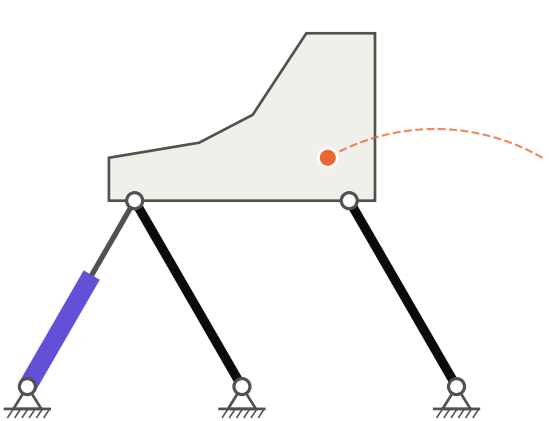
$\varphi = 60\,^\circ$, $s = 3.46$ m, $t = 0.00$ s



$\varphi = 90\,^\circ$, $s = 2.83$ m, $t = 2.54$ s



$\varphi = 120\,^\circ$, $s = 2.00$ m, $t = 5.86$ s